PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

TECHNICAL SPECIFICATION FOR DOC LEVELLER

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	03 (Incl, cover sheet)	Page 1 of 03

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	21/02/21

TITLE: TECHNICAL SPECIFICATION FOR DOC LEVELLER

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)		A4	A

TECHNICAL SPECIFICATION FOR DOC LEVELLER

1.	**Application**	a) To facilitate quick and safe loading / unloading of Goods by Forklift truck from inside the truck / container (viz. Finished lubes, Additive drums or Bags, Packaging materials on pallets or bags);
2.	**Quantity**	FP warehouse – 5nos Additive W/house – 3nos Packaging warehouse – 3nos Drum plant – 4nos Total – 15nos.
3.	**Design standard**	Conforming to European standard – EN-1398
4.	**Capacity**	Uniformly distributed load - 15MT Design Point load – 6MT
5.	**Model (VTC)**	**
6.	**Deck length**	3000mm
7.	**Deck Width**	2200mm
8.	**Frame height**	550mm – VTC
9.	**Lip Extension**	410mm – VTC
10.	**Bumpers**	Super Bumpers - to be included; Approx. size- 150mm width x 400mm height x 130mm depth including steel support;
11.	**Overall length (including Lip)**	3410mm (VTC)
12.	**Working height**	Maximum positive excursion - (+) 600mm Maximum negative excursion- (-) 300mm
13.	**Gradient**	Max. Upto 12.5% (upto 7 deg)
14.	**Powerpack**	Platform shall be safely and reliably controlled by a electro-hydraulic drive system of suitable rating with suitable safety features for safe operation & maintenance. Push-button to activate the dock leveler into operation; Electro-hydraulic power pack – to consist of all major components viz. Motor, High pressure Gear / Vane pump, Oil filters, Oil reservoir, solenoid valve, relief valve etc. all components of reputed make. Vendors to confirm the makes. Vendor to also confirm the location & space required for the Drive unit suitable for easy maintenance;

15.	**Operator & Control unit**	Electrohydraulic operator 400V, 50Hz, IP-55. Membrane push button with Press and Hold actuation and Emergency Stop integrated in IP-65 enclosure.
16.	**Platform**	Heavy duty, non-skid or anti-slip, chequered platform reinforced for maximum strength & rigidity;
17.	**Painting**	As per Manufacturer's standard; Structural members shall be preferably galvanized and then epoxy painted;
18.	**Safety features**	(a) Safety stop valve integrated in the lift cylinder; Lift cylinder to be of reputed make and with polyurethane seal of reputed make; (b) Maximum pressure valve © Maintenance crutch with positive engagement (d) Rubber bumpers to protect the dock levellers
19.	**Documentation**	Following documentation will be submitted on PO placement; - General Arrangement drawing - Maintenance manual - Electrical control system drawing - Inspection / Performance Test report for the unit - Guarantee certificate

PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

DATASHEET FOR GOODS LIFT MRL

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	19 (Incl, cover sheet)	Page 1 of 19

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	21/02/21

TITLE: DATASHEET FOR GOODS LIFT MRL

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)		A4	A

DATASHEET FOR GOODS LIFT MRL

1. SCOPE OF VENDORS:

Design, Engineering, preparation of detailed GA drawings for approval, manufacture / construction and testing, supply, installation, testing and commissioning including obtaining necessary approval from local authorities - for Goods Lift-MRL (Machine room less) – to be installed inside an production plant area, for movement of goods. The scope includes defects liability for a period of 1 year after completion of all works & handing over to client.

2. GENERAL

The equipment and installation covered by these specifications shall conform to codes of practice in force and highest standards of workmanship and materials. This work shall be done in accordance with the provisions of the Local Lifts Authority rules and shall also conform to requirements of local municipal by-laws and subsequent provisions, as also any state or local Act in force and latest Indian Standard 14665 and all latest applicable BIS, NBC code and 'CPWD General Specifications for Electrical Works (Part III, Lifts & Escalators) 2003'.

The Entire electrical installation shall be done in accordance with the Indian Electricity Act 2003, Indian Electricity Rules 1956 as amended to-date. The Electrical wiring shall strictly comply with IS:732 and latest applicable BIS and NBC code. The electrical works shall also conform to CPWD General Specification for Electrical Work Part-I (Internal) 1994 and Part-II (External) 1994 as amended up to date.

The Contractor shall follow all Statutory Requirements as well as best trade practices in the manufacture & installation of lifts. The Contractor shall arrange to obtain the statutory approval of the Inspectorate of Lifts as may be required for commissioning of the lifts and handover for operation after satisfactory tests.

3. DRAWINGS

The Contractor shall, submit 4(four) copies of all working drawings showing pit, hoistway layouts clearly indicating and specifying all connected structural, electrical and architectural works including imposed structural static / dynamic loads and electrical

ratings including calculations for selection of kW rating of motor. Within 10 days of receipt of letter of award of contract, the Contractor shall obtain from IOCL all the information he needs to prepare his drawings and finalise all parameters and data for design. The Contractor will be responsible for any discrepancies, errors and omissions in the drawings or particulars submitted by him even if these have been approved by the IOCL. On approval of these drawings (within 2 weeks of submission of full documentation), the Contractor shall submit 8(eight) copies of approved working drawings incorporating corrections / comments, if any, and shall immediately commence work.

On completion of work, the contractor shall supply four sets of CD's and 8 (eight) copies of the detailed wiring diagram, 'As built' drawings and equipment operation & maintenance manuals and original certificates from 'Inspector of Lifts' for all the lifts. The Contractor shall carry out all the work strictly in accordance with drawings, details and instructions of IOCL.

4. SOUND REDUCTION

The Contractor shall provide necessary sound reduction materials, such as rubber pads/ anti vibration pads of proper density to effectively isolate the machine from the machine beams and/or flooring.

5. TRACTION MACHINE

The machine shall be worm geared traction type with motor (steel worm, bronze gears, steel sheave shaft & Ferro molybdenum sheave), electro-mechanical type of brake and driving sheave mounted in proper alignment on a single heavy cast iron base or steel bedplate.

The worm shaft shall be fitted with roller bearings to take end thrust. The sheave shaft shall also be fitted with roller bearings to ensure proper alignment. All shafts shall be provided with well-designed keys.

Rotating parts shall be statically and dynamically balanced.

The drive sheave shall be designed with machined V-grooves to ensure adequate traction with minimum wear on rope. All sheaves including deflector sheaves, where used, shall conform to I. S. 14665 (Part 4 section 3)

Adequate and dust – proof lubrication shall be provided for all bearings and worm gears.

The brake shall be suitably curved and provided with fire proof friction lining. The operation of brake shall be smooth, gradual and with minimum noise. The brake shall be designed to be of adequate size and strength to stop and hold the car at rest with rated load. The brake shall be capable of operation automatically by various safety devices, current failure and by the normal stopping of the car. The brake shall be released electrically. It shall also be possible to release the brake manually so as to move the lift car in short stops. Suitable Brake release tools shall be supplied and stored in the machine rooms.

For manual operation of lifts, up & down direction of the movement of the car shall be clearly marked on the motor or traction machine. A warning plate in bold signal red colour to switch off the mains supply before releasing the brake and operating the wheel shall be prominently displayed.

6. HOIST MOTOR

The motor shall be suitable for 415 Volts +10% to –20%, 50 Hz. ± 5%, 3 Phase A.C. Supply. The motor must be designed for arduous lift duty, rapid reversals and constantly repeated starts & stops as defined in the relevant codes of practice. All windings must be heavily insulated, adequately impregnated for tropical climate and mechanically strengthened and must be specifically designed to have a high starting torque and low starting current characteristics within the limits acceptable to electricity supply co. requirements and I.E. Rules. The motor shall be designed in such a way as to withstand occasional overloading above its rated capacity and shall have overload protection. The motor shall have good speed regulation under different conditions of load and shall be designed to give a noiseless and vibration-free operation. Insulation shall be class F.

7. MOTOR CONTROL AND DRIVE

The lift motor shall be controlled by a variable voltage variable frequency (V.V.V.F.) micro-processor control system which shall control and monitor every aspect of lift operation at all stages of the car motion cycle on real time basis.

The A.C. V.V.V.F. drive system shall control A.C. voltage and frequency concurrently with the hoist motor to regulate the lift's actual performance to match closely the ideal speed pattern, obtain maximum efficiency of operation and provide a very smooth ride.

Frequency shall range fully between zero and rated value.

The Controller shall be provided with a self diagnostic programme to keep downtime to a minimum possible.

An Inspector's changeover test switch and set of test buttons shall be provided in the controller. Operation of the Inspector's changeover switch shall make both the car and landing buttons inoperative and permit the lift to be operated in either direction from machine room for test purposes by pressing corresponding test buttons in the controller. It shall not, however, interfere with the emergency stop switches inside the car or on the top of the car.

8. GUIDES AND FASTENINGS

i. Guide-rails for car and counterweight shall consist of machined mild steel Tee sections, erected plumb, and securely fastened to the lift well framing by heavy steel brackets, suitably spaced, to limit deflection of guide rails to 3 mm under normal working conditions.

ii. The guide-rails shall be of suitable section with ends tongued and grooved, forming matched joint and shall be connected with steel fish plates.

iii. Guide-rails shall cover the full height of the hoistway and pit , such that it shall not be possible for any of the car or counter weights shoes to run off the guides.

iv. Guides shall be designed to withstand the action of safety gear when stopping a counter weight or fully loaded car.

v. The max. deviation from true plumb and alignment of guide rails shall be 2 mm.

vi. All support framing shall be rigid and shall be designed to restrict displacement of the point of support of brackets to 3 mm under normal working conditions.

vii. The whole guide rail installation, including expansion joints, shall be designed for a smooth ride.

viii. The guide-rails shall be protected during storage and installation with a rust inhibiting coating which shall be cleaned off on completion of installation.

ix. Guide-shoes shall be adjustable type & mounted so as to provide continuous contact with guide rails under all conditions.

Guide shoes shall be provided at top and bottom of each side of car and counterweight and shall be designed for quiet operation.

Additional guide shoes shall be provided on each side of buffer frame in case of oil buffers.

Each lift shall be equipped with roller guides for up and down travel. There shall not be any metal-to-metal contact between Car and rail. Roller shall be mounted on ball bearings to provide quiet operation and excellent ride quality. (It is not required in case the design varies however the ride quality shall not be compromised for any other design).

9. SAFETY

In addition to other specifications, the lift shall be provided with safety devices as follows :-

i. Against overload

ii. Safety gear on car so that in the event of rope breaking or loosening, the car will be brought to rest immediately by means of grips on the guides.

The over speeding car shall be automatically brought to a gradual stop on guide rails and power supply to the hoist motor shall be switched off.

iii. Over speed centrifugal governor operating the safety gear in case of overspeeding of car in the down direction.

iv. Over travel limit switches at top and bottom limits of travel to disconnect the power supply and apply brakes to stop the car within a defined safe distance in case of over travel in either direction

v. Ultimate terminal switches to stop the car automatically within top & bottom clearances independently of normal over travel limit switches but with buffers operative.

vi. Protective guards to counterweights in pit, rope sheaves and wherever required.

vii. Toe guard apron to the car platform.

10. CAR

a. Cabin Size

The internal clear dimensions of the cabin shall not be less than those specified in IS 14665-Part I, NBC & CPWD General specifications for electric work (Lifts).

b. Frame and Safety Device

The car frame shall consist of mild steel channel/structural steel top and bottom securely riveted or bolted and substantially reinforced and braced so as to relieve the car enclosure of all strains when the safety device comes into action due to over speed or when the capacity loaded car is run on the buffer springs at normal speed.

The safety device mounted on the bottom members of the frame operated by a centrifugal speed governor shall be arranged to bring the car to a gradual stop on the guide rails in the event of excessive descending speed and provision shall be made to shut off the power supply to the motor.

c. Buffers

Substantial spring buffers (2 Nos.) shall be furnished and installed in the pit under the car and counterweight. These buffers shall be mounted on RCC Pedestals in the pit. The car buffer spring must be of correct design to sustain the

car with capacity load without damage should the car terminal limits become inoperative. The car buffers must be located symmetricaly with reference to centre of car.

The Contractor may alternatively offer oil type buffers. The plunger shall be mild steel, designed for a very high factor of safety and accurately machined. A toughened rubber bumper shall be fitted to the plunger top to cushion the impact of steel buffer plates attached under the car and the counterweight. An oil gauge shall be provided to check the oil level.

d. Counterweight

The lift shall be suitably counter-balanced for smooth and economical operation. Cast iron weights shall be contained in a structural steel frame properly guided with suitable guide shoes (minimum 4 Nos). It shall be equal to the total weight of lift plus approx. 50% of the contract load.

Substantial expanded metal counter-weight screen guard shall be furnished and installed at the bottom of hoist way, as required by Lift Inspector.

e. Hoisting and Governor Ropes

Bright steel wire ropes with fibre cores suitable for Lift duty as per BIS Code shall be used for hoisting ropes.

Not less than 3 independent suspension ropes shall be provided and designed to share load equally by means of adjustable shackle rods with equalizer springs at each end of hoisting ropes.

Each rope shall have adequate section to provide a minimum factor of safety of 4 based on the max. force on the rope.

Governor ropes shall be similar to hoisting ropes. Their ends shall be securely attached to the car and to the safety gear. The governor ropes shall be tensioned by a weight loaded device in the pit.

The contractor shall submit the technical details and source of supply of ropes to the IOCL as well as a certificate of performance of ropes from an approved test laboratory or Authority.

Compensation for travel shall be provided for all lifts having a travel of more than 30m.

f. Enclosure

The car enclosure shall be as specified in technical data sheet. The cabin floor, roof and walls shall be free of distortion and undue deflection as per IS 14665 – Part 4, Section 3.

g. Brakes

D.C. brakes will be spring-applied and electrically released. They shall be designed to provide smooth stops under variable loads.

h. Doors

Provision shall be made for vertical and horizontal fine adjustment of doors as per the specifications given in technical data sheet.

i. Door Operators

The door operators shall be VVVF inverter controlled heavy duty A. C. motor, allowing variable opening and closing speeds, and full synchronization of car and landing doors.

j. Travelling Cables

The traveling cables shall be multi-core with high conductivity stranded conductors specifically designed for lift duty. The cables shall be provided with retaining straps and individual cable clamps.

k. Emergency Stop Switch

A stop switch in the machine room / top of car shall be provided for use by maintenance crew to cancel all car and landing calls for a particular lift.

l. Maintenance Switch

On operation of the maintenance switch located on top of the car by the maintenance crew, the car shall travel at slow speed not exceeding 0.85 m / sec by continuous operation of a button

m. Landing Door Interlocks

Electrical interlocks shall be provided to ensure that the car does not operate unless all doors are closed and unless the car reaches a landing zone.

n. Overload Indicator

An overload indicator with buzzer shall be provided in the cabin to indicate that the car will not start as it is overloaded.

o. Other Features

All features specified in the BIS/NBC/CPWD and in the enclosed technical specifications shall be provided.

11. ELECTRIC WIRING

Necessary insulated wiring to connect all parts of the equipment shall be furnished and installed. Insulated wiring shall be flame retardant and moisture resistant and shall be run in G.S. conduits. All cables shall be flame – retardant with copper conductors.

Trailing cables shall be PVC sheathed copper conductor multi-core ribbon type designed for lift service and shall be flame retardant and moisture resistant. They shall be flexible and shall be suitably suspended to relieve strains on individual conductors. All copper conductors shall be of appropriate gauge copper to avoid excessive voltage drop. All

wires, cables, conduits, metal boxes, fittings and earthing shall comply with statutory requirements and BIS specifications.

The controller unit comprising of the MCCB, 25KA, adjustable overload and phase reversal and phase failure protection, all the circuit elements, transformer, rectifier for D.C. control supply, inverter power pack, terminal blocks etc. shall be enclosed in an insect vermin proof, sheet steel floor or wall mounted cabinet with hinged doors at front or at both front and rear. Proper warning boards and danger plates shall be provided on both sides of the controller casing. Sheet steel used for controller cabinet shall not be less than 14 gauge and shall be properly braced, where necessary. Suitable gland plate shall be provided for cable entry. The battery for the charger unit shall be suitably placed in the machine room. Degree of protection of Enclosure shall be IP54. Enclosure shall have provision of earthing studs.

All sheet steel work shall be painted with two coats of synthetic enamel paint of suitable shade both inside and outside over two coats of zinc primer.

Apart from lift controller enclosure, 7 distribution boards (3 Main DB + 4 DB) are required as per BOQ. Cables to incomer of these DB's shall be terminated by others, whereas outgoing cables for lift shall be in the scope of lift contractor. Contractor shall furnish the sizes of cables alongwith KW rating of motors.

12. PAINTING

All exposed metal work furnished in these specifications, except as otherwise specified, shall be given one shop coat of anti-corrosive EPOXY primer after approved surface treatment of metal surfaces and two coats of approved EPOXY /Polyurethane paint of approved shade. After installation of Lifts, a final Touch-up Coat of paint shall be applied.

13. WORKS TESTS

The following tests shall be carried out at Works. IOCL shall be given notice of the time and procedure of the tests before they are carried out, and shall be given facilities for observing the tests at Works.

a. High voltage works tests of equipment which is not already tested in accordance with appropriate IS codes.

b. Buffer test.

14. TESTS ON COMPLETION

The following tests shall be carried out to the satisfaction of the IOCL.

i. Insulation resistance and earth test for all electrical apparatus.

ii. Continuous operation of the lift under full load conditions and simulated starts and stops (150 nos. per hour each) for one hour at the end of which time the service temperature of the motor and the operating coils shall be tested. This shall be as per B.I.S. specification.

iii. The car shall be loaded until the weight on the rope is twice the combined weight of the car and the specified load. The load must be carried on for about 30 minutes, without any sign of weakness, temporary set or permanent elongation of the suspension rope strands.

iv. The following items shall be tested :

Levelling accuracy at each landing in conditions of fully loaded and empty car.
No load current and voltage readings both on 'Up' and 'Down' Circuits.
Full load current and voltage readings both on 'Up' and 'Down' Circuits.
One and quarter load current and voltage readings both on 'Up and 'Down' Circuits.
Stalling current and voltage and time taken to operate overload.
Overload protection.
Gate sequence relays, if provided and installed.
Car and landing door interlocks.
Collective control and priority sequences, if installed.
Safety gear mechanism for car and counterweight with fully loaded car and also with only 68 kg load.
Speeds on Up and Down travel with full load, half load and empty car.
Door contacts.

Final terminal stopping device.
Normal terminal stopping device.
Car and counterweight buffers with contract load and contract speed.
Operation of controllers.
Manual operation of lift at mid-way travel.
Emergency operation.

v. Tests on completion shall also be performed to the satisfaction of Inspector of Lifts and a certificate will be obtained from the 'Lift Inspector ' by the contractor.

15. STATUTORY APPROVALS

All statutory approvals from commencement to commissioning of lifts shall be obtained by the Contractor from the Inspector of Lifts and / or other authorities. However, the client will provide all necessary assistance for providing documents, drawings and certificates pertaining to other contractors, if required.

The contractor shall pay necessary fees in connection with the approval of installation of lifts.

16. FEATURES REQUIRED FOR VVVF LIFTS

(a) **Fireman's Switch**

A fireman's toggle switch shall be provided in a break glass for the specified lift at ground floor to enable firemen to bring the lift non-stop to ground floor from any location and to cancel all calls until the car is operated on attendant control.

(c) **Emergency Power Operation**

In case of power failure, standby power equipment shall enable lifts to reach a pre-determined floor and then permit operation of one or more lifts on emergency power.

A trickling battery shall be provided to supply power to light fixtures, fan, alarm and intercom.

(d) **Profile Generator**

A profile generator or similar device shall be provided to use the car at an optimum speed level and to improve levelling accuracy.

(f) **Home Landing Facility**

A car shall return to a pre-determined landing after the last call is answered.

(g) **Door Safety**

Multi-beam infrared / ultrasonic electronics curtains shall be provided to scan the doorway and reverse the door closing in case of any obstruction.

(k) **Manual Cranking & Slow speed Travel**

A manual cranking facility shall be provided.

Slow speed operation shall be possible from machine room and car top.

(l) **Auto Fan Off**

In case no calls are registered for a pre-set time, the cabin fan shall be automatically switched off.

(m) **Automatic Rescue Device**

In case of mains power failure and Lift control system failure, the Lift's own rechargeable and maintenance free battery power shall move the car to the nearest floor and the door shall open automatically. A battery run-down indicator shall be provided.

17. PERFORMANCE PARAMETERS

The following parameters shall be achieved in the installation :

*	Levelling Accuracy	± 3 mm for 1.5 m/s speed ± 4 mm for 0.75 m/s speed
*	Jerk level	0.9 – 1.5 m/s^3
*	Noise level in car	58 dB
*	Noise level at 1 M in machine room	60 dB
*	Acceleration rate	0.6 – 1.0 m/s^2 (adjustable)
*	Max. car vibration	20 milli gals.

18. SUBMITTALS 'ALONGWITH TENDER' AND 'POST AWARD'

(A) The following items are required to be submitted in duplicate **along with the Tender**.

i. Catalogues with offered items highlighted.
ii. List of imported components, if any.
iii. Compliance Statement for guaranteed performance parameters given in Specification 19.0 above.
iv. Confirmation that offer submitted meets the technical specifications & scope of work and there are no deviations and exclusions from Tender Specifications.
v. The contractor shall specify in his offer the full capability of his system in this regard.

(B) The successful contractor, **after award of the contract** ,shall furnish following technical particulars of the equipment/devices for the approval by IOCL.

i) Single line/ schematic diagram of electronic control panel, lift & equipment etc.
ii) Layout of Hoistway, showing foundation details in the pit, machine room, electric control panel, Lift & equipment etc.
iii) Earthing layout.
iv) Inspection manual for equipment & accessories covered in the scope of supply (8 copies).

v) Technical literature of operation, control and maintenance etc. (8 copies) alongwith CDS.

vi) Schedule of scope of maintenance service during defect liability period and AMC.

The technical parameters furnished by the tenderer would be examined in detail during design submission stage. All improvements considered necessary to meet the tender Technical Specifications would have to be incorporated without any additional cost to IOCL with objective of providing high performance and safety Lifts.

19. MAINTENANCE DURING DEFECTS LIABILITY PERIOD

Comprehensive maintenance during Defects Liability Period inclusive of periodic servicing, prompt attention to client complaint, prompt rectification of all malfunctions and equipment failures, replacement of defective equipment / parts, replacement of light fittings, lubrication including lubricants, maintaining correct alignment and levelling of cars and ensuring smooth running, starts and stops etc. all complete to IOCL's satisfaction shall be done.

DATASHEET FOR GOODS LIFT (MRL)

1.	Quantity	Three Nos.
2.	Capacity	2000 kgs
3.	Application	For installation inside our – for movement of packaging materials from GF to First and Mez. Floor;
4.	No. of stops / openings	03 stops / 03 Openings / Ground, First and Mezz. flr.
5.	Travel distance	15 Mtrs
6.	No. of Stops	3 nos
7.	Hoistway Size	VTA
8.	Car Inside dimensions	2800 mm Depth X 2200 mm Width X 2200 mm Ht. (VTC)
9.	Clear Opening on Shaft	2500 mm Width X 2000 mm Height
10.	Speed	0.50 m/s (VTC)
11.	Car Panels	C.S. Powder coated Finish Cabin
12.	Car Flooring	Aluminum chequered plate flooring
13.	Car Entrance	MS collapsible Door
14.	Landing Entrance	MS collapsible Door
15.	Overhead Available	2000 mm
16.	Pit Depth Available	Not available;
17.	Drive	VVVF with self-braking Geared motor
18.	Control	Microprocessor Based Down Collective With Soft Stop valve
19.	Signals	Position Indicators / Direction Arrows / Arrows with Scrolling
20.	Limit Switches	Limit switches for Over travel, Cabin entrance & exit doors
21.	Emergency Brake release device	Manual Brake release device required for emergency descent;
22.	Emergency Light and Alarm arrangement	Required

23.	Entry / Exit Pattern	Multiple entry/exit pattern
24.	Guide Rails	Machined, T-section rails of rolled mild steel having mirror finish.
25.	Suspension Ropes	ISI Marked, self lubricating type for car. USHA MARTIN or equivalent make;
26.	Safeties	Suitable Safety features to be provided to control the speed of the Lift and ensure operational safety of the system. These may include amongst others :
		a) Electromechanical type Gate locks for landing doors to ensure proper locking of the door before starting of the Lift.
		b) Car Gate Switch to be provided for the car door to enable running of the Lift only after proper closing of car door.
		c) Mechanical type Final Limit Switch to cut off the supply of the Controller in case of over running of the Lift.
		d) Earthing to be provided to the Car, Gate locks, Push button box, Guide rails to ensure electrical safety.
27.	Controllers	Suitable controllers shall be provided for trouble free operation and ease of maintenance;
28.	Wiring & Cabling	All wiring and cabling in the shaft way will be passed through metallic cable trays with requisite earthing to ensure safety of electrical Installation of the system.
29.	Buffer Spring	As an extra precautionary measure a pair of springs of appropriate compressibility, known as buffers are installed at the bottom of the shaft way to prevent the car from landing with a thud in case of snappage of suspension ropes, this, of course is a rare posssibility.
30.	Swing Doors at each Exit with electronic lock	Required
31.	Provision for Lockout Master switch	Required
32.	Overspeeding governor in case of wire rope failure	Required

33.	Operating Area classification	Safe Area;
34.	Power Supply	440 Volts, 50 Hz, A C Supply
35.	Light Supply	230 Volts, 50 Hz, A C Supply

20. CONTROL SYSTEM

The Goods Lift shall be supplied with all safeties & controls required for the safe operation of the unit. The Area Classification to be considered – is SAFE AREA.

21. INSTALLATION SERVICES & TRAINING

Testing & commissioning of system shall be included in the offer. The scope also includes the complete training of the plant operating personnel in the Operations & Maintenance of the system. The training shall be done immediately after commissioning of the Goods lift. A comprehensive operating and maintenance manual in soft and hard form shall also be provided by the vendor.

22. DOCUMENTATION

With Quotation

A. General Arrangement Drawings
B. Catalog / Manuals for various components

On finalization of PO :

A. General Arrangement Drawings for approval
B. Assembly installation Drawing
C. Specifications of all major bought out components
E. Installation & Maintenance manual
F. Inspection & Test reports
G. Guarantee / Warranty certificates
H. Approval from local authorities

Enclosure

Plant Layout Plan

PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

DATASHEET FOR TELESCOPIC BELT CONVEYOR

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	04 (Incl, cover sheet)	Page 1 of 04

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	21/02/21

TITLE: DATASHEET FOR TELESCOPIC BELT CONVEYOR

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)		A4	A

DATASHEET FOR TELESCOPIC BELT CONVEYOR

Sr.No.	ITEMS	Description
	SCOPE	Design, Engineering, drawing approval, procurement, fabrication/manufacture, assembly, testing, supply, installation assistance and commissioning of Telescopic Belt conveyors
	SERVICE	For use in : a) Packaging material Warehouse – for unloading of packaging materials from Trucks for storage in Warehouse (viz. Flat Cartons, Stacks of Pails, Carton boxes of Caps etc.) - 1no. b) Finished product warehouse – for Loading of FP cartons for small packs / Filled pails etc. into Trucks – 2nos The units are for Indoor application but due to the coastal weather conditions – environment is highly corrosive in nature; Painting specifications to consider these factors; To consider highly corrosion resistant Epoxy or equivalent paint of required DFT.
	PROJECT SITE	Amullavoyal village at Chennai . Tamil Nadu State The requirement is for a Greenfield, Grassroot Lube Complex – being set up at Chennai;
1.	Quantity	3 Nos.
2.	Material To Be Handle	Boxes / Flat Carton bundles / Stacks of Pails etc. as mentioned above.
3.	Weight Of The Box/Bags	25-40 Kg. Maximum
4.	Size Of Box/Bags	600 (L) x 500 (W) x 600 (H) (OVERALL)
5.	Temperature	Ambient
6.	Capacity	550 Boxes Per Hour
7.	Type	Collapsible Boom Type Hinged Telescopic Belt Conveyor;
8.	Design Capacity	600 Per Hour
9.	Size Of Belt	750 mm Width

10.	Type Of Belt	Rough Top 2 Ply, PVC Coated Endless Type
11.	Length Of Conveyor	Fixed Length : 4800 mm Telescopic (Expanded) : 6000 mm Total : 10,800 mm
12.	Main Body Of Conveyor	Manufacturing From IS 226 Plate Material - VTC
13.	Snub Pulley for Conveyor	VTC
14.	Drive Pulley	VTC
15.	Telescopic Movement	Mechanical Arrangement
16.	Mounting	On Fixed MS Chassis Mounted On Fabricated Frame with FIX and CASTOR Wheel with Manual Toeing Bar
17.	Height Adjustment	With 2 HP Power Pack & Cylinder for Height Adjustment (All components to be of reputed make - VTC makes of components)
18.	Panel	The Conveyor comes with a Fixed Main Panel on Rear Side and Front Side Only; Operating Panel with Emergency Stopper Sensor Bar, LED Lighting, Totally Consoled Wiring.
19.	Painting	INTERNAL : High corrosion resistant Epoxy paint of required DFT as per Vendor standard; VTC;
	DRIVE FOR BELT	**SPECIFICATION**
1.	Belt Motor	** HP, 1440 RPM, TEFC – VTC details of Make, type etc.
2.	Motor Make	Rotomotive or suitable equivalent make – VTC
3.	Gear Box	Worm Type, Wall Mounted , With Suitable Ratio - VTC
4.	Gear Make	Rotomotive or suitable equivalent make - VTC
	DRIVE FOR TELE. MOVEMENT	**DESCRIPTION**

1.	Motor	** HP, 1440 RPM, TEFC- VTC make, HP and Type;
2.	Motor Make	Rotomotive or suitable reputed equivalent make – VTC
3.	Gear Box	Worm Type, Wall Mounted , With Suitable Ratio – VTC
4.	Gear Make	Rotomotive or equivalent – VTC
	DRIVE FOR TRANSVERSE MOVEMENT	**CROSS TRANSVERSE LENGTH 10.8 MTR**
1.	Motor	** HP – VTC make, HP, RPM and Type;
2.	Motor Make	Rotomotive or Equivalent – VTC
3.	Gear Box	Worm Type, Wall Mounted , With Suitable Ratio – VTC
4.	Gear Make	Rotomotive or suitable reputed equivalent make - VTC
	ACCESSORIES	The unit shall be complete with all necessary accessories – required for normal operation & Maintenance of the machine; To ensure safety of personnel at the time of working suitable arrangements to be provided in the design, in such a manner and location as necessary to give safe and complete access to all the parts of the machine requiring inspection or adjustments.
	DOCUMENTATION	Vendor to confirm that following documentation will be submitted in 3 sets on placement of PO: -General Arrangement drawing - Installation drawing -Equipment layout and static loads - Guarantee / Warranty Certificate -Manufacturer's original Test Certificates & Guarantee certificates for bought out components - Inspection / Performance Test report for the unit - Installation manual - Operations & Maintenance manual

PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

TECHNICAL SPECIFICATION FOR FORK LIFT TRUCK

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	03 (Incl, cover sheet)	Page 1 of 03

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	21/02/21

TITLE: TECHNICAL SPECIFICATION FOR FORK LIFT TRUCK

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)		A4	A

TECHNICAL SPECIFICATION FOR FORK LIFT TRUCK

1.	**Application**	a) For Movement of Finished product –on Pallets from Filling Hall to FP Warehouse;
		b) For Loading of FP from FP Warehouse into Trucks /Containers;
		c) For Unloading & movement of Additive Drums /Bags on pallets to Additive Warehouse;
		d) For Unloading & movement of empty Packaging materials to empty Packaging warehouse;
2.	**Quantity**	Approx. 10nos. with breakdown as below (to be reviewed);
3.	**Type**	Electric (Battery operated) Forklift truck- 6 nos Diesel forklift trucks – 4nos;
4.	**Capacity**	2 MT capacity at 500mm load Centre
5.	**Model (VTC)**	
6.	**Forklength**	1000mm long forks
7.	**Features**	- Jerk free handling of Loads - Head & tail lights - Load back rest - Standard tool kit - Suitable Battery (Vendors to give detail) - Power Steering - 3 Stage clear view full free lift Mast
8.	**Maximum Lift Height**	6000mm (VTC)
9.	**Collapsed Height**	VTC
10.	**Free lift**	2400mm
11.	**Accessories**	- Battery Charger – 3ph, 440 V, 12 hour Recharge time rating - Battery Discharge Indicator with Lift lock out
10.	**Documentation**	Following documentation will be submitted in 3 sets: - General Arrangement drawing - Maintenance manual - Electrical control system drawing - Inspection / Performance Test report for the unit - Guarantee certificate

PEC LTD.
21 Shipyard Road, Singapore 628 144
Tel: 6268 9788 Fax: 6268 9488
URL: http://www.peceng.com

DISTRIBUTION	
INTERNAL	
HEAD OFFICE	
SITE OFFICE	
EXTERNAL	
CLIENT	1
CONSULTANT	
SUB CON .1	
SUB CON .2	
SUB CON .3	
TOTAL	1
PURPOSE	
INFORMATION	
PRELIMINARY	
TENDER	√
COMMENT	
REVIEW	
APPROVAL	
CONFIRMATION	
CONSTRUCTION	
AS-BUILT	

TECHNICAL SPECIFICATION FOR BATTERY OPERATED PALLET TRUCK

Project:

GRASSROOT LUBE COMPLEX PROJECT AT AMMULAVAYOL, CHENNAI

Consultant:

PEC LTD.

Client:

INDIAN OIL CORPORATION LIMITED

	Total No. of Sheets	Page
	02 (Incl, cover sheet)	Page 1 of 02

REV.	DESCRIPTION	PREP BY	CHK BY	TECH APP	PROJ APP	DATE
A	ISSUED FOR TENDER	HTO	SBS	DR	KGK	21/02/21

TITLE: TECHNICAL SPECIFICATION FOR BATTERY OPERATED PALLET TRUCK

PEC PROJECT NO.	EPS JOB NO.	IOC JOB NO.	DOCUMENT NO.	SIZE	REV.
0RG2N0410000	0RG2N0410000	HCC/LUBES-07/GT-09/2018-19 (E-Tender ID: 2018_MKTHO_80987_1)		A4	A

TECHNICAL SPECIFICATION FOR BATTERY OPERATED PALLET TRUCK

1.	**Application**	a) For Movement of Additive / Raw Material –on Pallets from Warehouse to Process;
		b) For Movement of empties /Packaging materials from Warehouse to Filling Hall
2.	**Quantity**	Approx. 6nos.
3.	**Type**	Powered Pallet trucks suitable for Pallets 1200 x 1200mm
4.	**Capacity**	2 MT capacity at 600mm load Centre
5.	**Model (VTC)**	Model including Operator Platform
6.	**Forklength**	1000mm long forks
7.	**Features**	- Jerk free handling of Loads - Head & tail lights - Standard tool kit - Suitable Battery (Vendors to give details)
8.	**Maximum Lift Height**	6000mm (VTC)
9.	**Collapsed Height**	VTC
10.	**Accessories**	- Battery Charger – 3ph, 440 V, 12 hour Recharge time rating - Battery Discharge Indicator with Lift lock out
11	**Documentation**	Following documentation will be submitted on PO placement: - General Arrangement drawing - Maintenance manual - Electrical control system drawing - Inspection / Performance Test report for the unit - Guarantee certificate